RESEARCH PROJECT CONCEPT · MIDTERM

Highlighted: components this project adds

# Negation-Aware Retrieval for Russian and Kazakh Legal Texts

**Research question:** do a morphology-aware BM25 and an NLI reranker retrieve the article with the right polarity, and do they help Kazakh more than Russian?

**Query**
legal question in Ru or Kz

→

**Hybrid search**
e5 dense + BM25 that keeps negation

→

**RRF fusion**
merges both rankings

→

**NLI reranker**
XLM-R: entails vs contradicts

→

**Top-5 articles**
LLM answer on top: optional

## Problem

On NevIR, most dense retrievers score 6–23% on negation pairs, below random (25%). Then RAG states the opposite of the law. Matters for lawyers, citizens, legal tech.

## Approach

Ru and Kz versions of the same articles of the Labor and Tax Codes. Add one part at a time: dense, + BM25, + morphology, + NLI reranker; a plain reranker and a rule as controls.

## Hypotheses

**D1** e5 puts negation closer than a paraphrase
**D2** the tokenizer rarely isolates Kz -ма/-ме
**H1** morph. BM25 shrinks the Negation Gap
**H2** NLI beats a plain reranker and a rule
**H3** bigger gain for Kz than Ru, same articles
**H5** Turkish XNLI improves the Kz reranker
**H6** “не менее” examples cut false contradictions

## Significance

Kazakh marks negation with an affix (төле-ме-йді) that BPE can split away. No Kazakh legal retrieval benchmark and no established Kazakh NLI dataset.

## Contribution

To our knowledge, the first Ru/Kz legal retrieval benchmark with negation queries, own diagnostics and an ablation. Hybrid search and reranking already exist.

## Own metrics

**Negation Margin** cos(paraphrase) − cos(negation)
**Affix Isolation** Kz negative forms split at the affix
**Twin-above-Gold** opposite article above the right one
**Negation Gap** Recall@5 general minus negation

**Feasibility:** two codes, base-size models on one GPU. Main risk: manual labeling of test questions (fallback ≈90 queries).